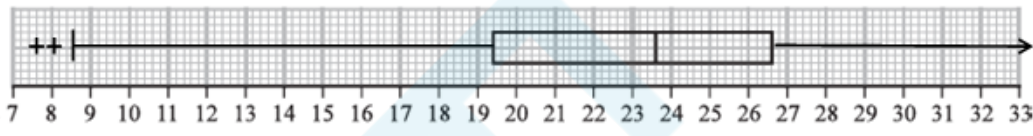


Mark Scheme

Q1.

Part	Working or answer an examiner might expect to see	Mark	Notes
(a)	$IQR = 2.6 - 19.4 = 7.2$	B1	This mark is given for finding the interquartile range
	$19.4 - (1.5 \times 7.2) = 8.6$ $19.4 + (1.5 \times 7.2) = 37.4$	M1	This mark is given for a method find the values for the whiskers of the boxplot
			
		A1	This mark is given for plotting the correct whisker (8.6) on the boxplot
		A1	This mark is given for plotting the two correct outliers 7.6 °C and 8.1 °C
(b)	October (since it is the month with the coldest temperatures between May and October in Beijing)	B1	This mark is given for a correct suggestion with a supporting reason.
(c)	$\sigma = \sqrt{\frac{S_{xx}}{n}} = \sqrt{\frac{4952.906}{184}} = \sqrt{26.92} = 5.19$	B1	This mark is given for showing the calculation for the standard deviation to three significant figures
(d)	$z = (\pm) 1.2816$	B1	This mark is given for identifying the z-value for the 10th and 90th percentiles (from tables or calculator)
	$2 \times z \times 5.19$	M1	This mark is given for a method to find the interpercentile range between the 10th and 90th value
	$= 13.303$	A1	This mark is given for finding a correct interpercentile range between the 10th and 90th value
(e)	Daily wind speed (Beaufort) since it is qualitative data	B1	This mark is given for stating a correct variable with a supporting reason
	Rainfall (since it is not symmetric)	B1	This mark is given for stating a correct variable with a supporting reason
(Total 11 marks)			

Q2.

Part	Working or answer an examiner might expect to see	Mark	Notes
(a)	$\frac{24.63 - 25}{\sigma} = -1.0364$	M1	This mark is given for standardising as part of a method to find σ
	$\sigma = 0.357$	A1	This mark is given for a correct value of σ
	$P(D > K) = 0.4$ or $P(D < K) = 0.6$	B1	This mark is given for
	$\frac{k - 25}{\sigma} = \frac{k - 25}{0.357} = 0.2533$	M1	This mark is given for using a normal model to find the probability
	$k = 25.09$	A1	This mark is given for a correct value for k
(b)	$Y \sim B(200, 0.45)$ so $W \sim N(90, 49.5)$	B1	This mark is given for setting up the normal distribution approximation of the binomial
	$P(Y < 100) \approx P(W < 99.5) = P\left(Z < \frac{99.5 - 90}{\sqrt{49.5}}\right)$	M1	This mark is given for using the normal model with a continuity correction
	$= 0.912$	A1	This mark is given for finding a correct value of the probability
(c)	$H_0 : \mu = 25$ $H_1 : \mu < 25$	B1	This mark is given for both hypotheses in terms of μ found correctly
	$\bar{D} \sim N\left(25, \frac{0.16^2}{20}\right)$	M1	This mark is given for a method to set up the normal distribution
	$P(\bar{D} < 24.94) = 0.0468$	A1	This mark is given for using the model to find a correct p -value
	$p = 0.0468 < 0.05$, so reject H_0	M1	This mark is given for a correct comparison and non-contextual conclusion
	There is sufficient evidence to support Hannah's belief	A1	This mark is given for a correct conclusion in context stated
(Total 13 marks)			

Q3.

Question	Scheme	Marks	AOs
(a)	Area = $8 \times 1.5 = 12 \text{ cm}^2$ Frequency = 8 so $1 \text{ cm}^2 = \frac{2}{3} \text{ hour (o.e.)}$	M1	3.1a
	Frequency of 12 corresponds to area of 18 so height = $18 \div 2.5 = 7.2 \text{ (cm)}$	A1	1.1b
	Width = $5 \times 0.5 = 2.5 \text{ (cm)}$	B1cao	1.1b
		(3)	
(b)	$[\bar{y}] = \frac{205.5}{31} = \text{awrt } 6.63$	B1cao	1.1b
	$[\sigma_y] = \sqrt{\frac{1785.25}{31} - \bar{y}^2} = \sqrt{13.644641} = \text{awrt } 3.69$	M1	1.1a
	allow $[s] = \sqrt{\frac{1785.25 - 31\bar{y}^2}{30}} = \text{awrt } 3.75$	A1	1.1b
		(3)	
(c)	Mean of Heathrow is higher than Hum and standard deviation smaller suggesting Heathrow is more reliable	M1	2.4
	Hum is South of Heathrow so does <u>not</u> support his belief	A1	2.2b
		(2)	
(d)	$\bar{x} + \sigma \approx 10.3$ so number of days is e.g. $\frac{(11 - "10.3")}{3} \times 8 (+5)$	M1	1.1b
	= 6.86 so 7 days	A1	1.1b
		(2)	
(e)	$[H = \text{no. of hours}] \quad P(H > 10.3) \text{ or } P(Z > 1) = [0.15865 \dots]$	M1	3.4
	Predict $31 \times 0.15865 \dots = \underline{\underline{4.9 \text{ or } 5 \text{ days}}}$	A1	1.1b
		(2)	
(f)	(5 or) 4.9 days < (7 or) 6.9 days so model may not be suitable	B1	3.5a
		(1)	
(13 marks)			

Notes:	
(a)	
M1:	for clear attempt to relate the area to frequency. Can also award if their height $\times$ their width = 18
A1:	for height = 7.2 (cm)
(b)	
M1:	for a correct expression for σ or s , can ft their value for mean
A1:	awrt 3.69 (allow $s = 3.75$)
(c)	
M1:	for a suitable comparison of standard deviations to comment on reliability.
A1:	for stating Hurn is south of Heathrow and a correct conclusion
(d)	
M1:	for a correct expression – ft their $\bar{x} + \sigma \approx 10.3$
A1:	for 7 days but accept 6 (rounding down) following a correct expression
(e)	
M1:	for a correct probability attempted
A1:	for a correct prediction
(f)	
B1:	for a suitable comparison and a compatible conclusion

Q4.

Question	Scheme EXAM PAPERS PRACTICE	Marks	AOs
(a)			
	$P(L > 50.98) = 0.025$	B1cao	3.4
	$\therefore \frac{50.98 - \mu}{0.5} = 1.96$	M1	1.1b
	$\therefore \mu = 50$	A1cao	1.1b
	$P(49 < L < 50.75)$	M1	3.4
	$= 0.9104\dots$ awrt <u>0.910</u>	A1ft	1.1b
		(5)	
(b)	$S = \text{number of strips that cannot be used so } S \sim B(10, 0.090)$	M1	3.3
	$= P(S \leq 3) = 0.991166\dots$ awrt 0.991	A1	1.1b
		(2)	
(c)	$H_0 : \mu = 50.1 \quad H_1 : \mu > 50.1$	B1	2.5
	$\bar{X} \sim N\left(50.1, \frac{0.6^2}{15}\right)$ and $\bar{X} > 50.4$	M1	3.3
	$P(\bar{X} > 50.4) = 0.0264$	A1	3.4
	$p = 0.0264 > 0.01$ or $z = 1.936\dots < 2.3263$ and not significant	A1	1.1b
	There is insufficient evidence that the <u>mean length</u> of strips is <u>greater than 50.1</u>	A1	2.2b
		(5)	
(12 marks)			

Notes:

(a)

1st M1: for standardizing with μ and 0.5 and setting equal to a z value ($|z| > 1$)

2nd M1: for attempting the correct probability for strips that can be used

2nd A1ft: awrt 0.910 (allow ft of their μ)

(b)

M1: for identifying a suitable binomial distribution

A1: awrt 0.991 (from calculator)

(c)

B1: hypotheses stated correctly

M1: for selecting a correct model (stated or implied)

1st A1: for use of the correct model to find $p = \text{awrt } 0.0264$ (allow $z = \text{awrt } 1.94$)

2nd A1: for a correct calculation, comparison and correct statement

3rd A1: for a correct conclusion in context mentioning "mean length" and 50.1

Q5.

Question	Scheme	Marks	AOs
(a)	The seeds would be destroyed in the process so they would have none to sell	B1	2.4
		(1)	
(b)	$[S = \text{no. of seeds out of 24 that germinate, } S \sim B(24, 0.55)]$		
	$T = \text{no. of trays with at least 15 germinating. } T \sim B(10, p)$	M1	3.3
	$p = P(S \geq 15) = 0.299126\dots$	A1	1.1b
	So $P(T \geq 5) = 0.1487\dots$ awrt <u>0.149</u>	A1	1.1b
		(3)	
(c)	n is large and p close to 0.5	B1	1.2
		(1)	
(d)	$X \sim N(132, 59.4)$	B1	3.4
	$P(X \geq 149.5) = P\left(Z \geq \frac{149.5 - 132}{\sqrt{59.4}}\right)$	M1	1.1b
	$= 0.01158\dots$ awrt <u>0.0116</u>	A1cso	1.1b
		(3)	
(e)	e.g The probability is very small therefore there is evidence that the company's claim is incorrect.	B1	2.2b
		(1)	
(9 marks)			

Notes:
(a) B1: cao
(b) M1: for selection of an appropriate model for T 1 st A1: for a correct value of the parameter p (accept 0.3 or better) 2 nd A1: for awrt 0.149
(c) B1: both correct conditions
(d) B1: for correct normal distribution M1: for correct use of continuity correction A1: cso
(e) B1: correct statement

Q6.

Qu	Scheme	Marks	AO
(a)	$P(L > 16) = 0.69146...$ awrt 0.691	B1 (1)	1.1b
(b)	$P(L > 20 L > 16) = \frac{P(L > 20)}{P(L > 16)}$ $= \frac{0.308537...}{(a)} \text{ or } \frac{1-(a)}{(a)} = 0.44621...$ For calc to work require $(0.44621...)^4 = 0.03964...$ awrt 0.0396	M1 A1ft A1 dM1 A1 (5)	3.1b 1.1b 1.1b 2.1 1.1b
(c)	Require: $[P(L > 4)]^2 \times [P(L > 20 L > 16)]^2$ $= (0.99976...)^2 \times (0.44621...)^2$ $= 0.19901...$ awrt 0.199 (*)	M1 A1ft A1cso* (3)	1.1a 1.1b 1.1b
(d)	$H_0: \mu = 18 \quad H_1: \mu > 18$ $\bar{L} \sim N\left(18, \left(\frac{4}{\sqrt{20}}\right)^2\right)$ $P(\bar{L} > 19.2) = P(Z > 1.3416...) = 0.089856...$ (0.0899 > 5%) or (19.2 < 19.5) or 1.34 < 1.6449 so not significant Insufficient evidence to support Alice's claim (or belief)	B1 M1 A1 A1 A1 (5)	2.5 3.3 3.4 1.1b 3.5a
(14 marks)			

Notes	
(a)	B1 for evaluating probability using their calculator (awrt 0.691) Accept 0.6915
(b)	1 st M1 for a first step of identifying a suitable conditional probability (either form) 1 st A1ft for a ratio of probabilities with numerator = awrt 0.309 or 1 - (a) and denom = their (a) 2 nd A1 for awrt 0.446 (o.e.) Accept 0.4465 (from $\frac{0.3085}{0.691} = 0.44645...$) NB $\frac{P(16 < L < 20)}{P(L > 16)} = 0.5538...$ scores M1A1A1 when they do $1 - 0.5538 = 0.4462...$ 2 nd M1 (dep on 1 st M1) for 2 nd correct step i.e. (their 0.446...) ⁴ or $X \sim B(4, "0.446")$ and $P(X = 4)$ 3 rd A1 for awrt 0.0396
(c)	1 st M1 for a correct approach to solving the problem (May be implied by A1ft) 1 st A1ft for $P(L > 4) =$ awrt 0.9998 used and ft their 0.44621 in correct expression If use $P(L > 20) = 0.3085...$ as 0.446.. in (b) then M1 for $(0.3085...)^2 \times [P(L > 4)]^2$; A1ft as above * 2 nd A1cso for 0.199 or better with clear evidence of M1 [NB $(0.4662...)^2 = 0.199...$ is M0A0A0] Must see M1 scored by correct expression in symbols or values (M1A1ft)
(d)	B1 for both hypotheses in terms of μ . M1 for selecting a suitable model. Sight of <u>normal</u> , <u>mean</u> 18, <u>sd</u> $\frac{4}{\sqrt{20}}$ (o.e.) or <u>variance</u> = 0.8 1 st A1 for using the model correctly. Allow awrt 0.0899 or 0.09 from correct prob. statement CR $(\bar{L}) > 19.471...$ (accept awrt 19.5) or CV of 1.6449 (or better: calc 1.6448536..)
ALT	2 nd A1 for correct non-contextual conclusion. Wrong comparison or contradictions A0 Error giving 2 nd A0 implies 3 rd A0 but just a correct contextual conclusion can score A1A1 3 rd A1 dep on M1 and 1 st A1 for a correct contextual conclusion mentioning <u>Alice's claim</u> / <u>belief</u> or there is insufficient evidence that the mean <u>lifetime</u> is more than 18 hours

Q7.

	Scheme	Marks	AO
(a)	$\left[\text{Let } F \sim N(166.5, 6.1^2) \right] P(F < k) = 0.01 \Rightarrow \frac{k - 166.5}{6.1} = -2.3263$ $k = 152.309... \quad \underline{152} \text{ or awrt } \underline{152.3}$	M1 A1 (2)	3.4 1.1b
(b)	$[P(150 < F < 175) =] \quad 0.914840... \quad \text{awrt } \underline{0.915}$	B1 (1)	1.1b
(c)	$P(F > 160 150 < F < 175)$ $= \frac{P(160 < F < 175)}{P(150 < F < 175)} \text{ or } \frac{P(160 < F < 175)}{"(b)"} = \frac{0.7749487...}{0.914840...} = 0.84708... \text{ awrt } \underline{0.847}$	M1 M1 A1ft A1 (4)	3.1b 1.1b 1.1b 1.1b
(d)	$H_0: \mu = 166.5 \quad H_1: \mu < 166.5$ $[\text{Let } \bar{X} = \text{height of female from 2}^{\text{nd}} \text{ country}] \quad \bar{X} \sim N\left(166.5, \left(\frac{7.4}{\sqrt{50}}\right)^2\right)$ $P(\bar{X} < 164.6) = 0.03472...$ $[0.0347... < 0.05 \text{ so significant or reject } H_0]$ There is evidence to support Mia's belief	B1 M1 A1 dA1 (4)	2.5 3.3 3.4 2.2b
(11 marks)			

	Notes
(a)	M1 for standardising (allow $\pm$) with k , 166.5 and 6.1 and set equal to a z value $2.3 < z < 2.4$ A1 for 152 or awrt 152.3 Ans only 2/2 [Condone poor use of notation e.g. $P(\frac{k-166.5}{6.1}) = -2.3263$] Allow percentages instead of probabilities throughout.
(b)	B1 for awrt 0.915
(c)	1 st M1 for interpreting demand as an appropriate conditional probability ($\Rightarrow$ by 2 nd M1) 2 nd M1 for correct ratio of expressions (can fit their (b) on denominator) ($\Rightarrow$ by 1 st A1ft) 1 st A1ft for a correct ratio of probs (can fit their "0.9148..." to 3sf from (b) if > 0.775) 2 nd A1 for awrt 0.847
(d)	B1 for both correct hypotheses in terms of μ 1 st M1 for selecting the correct model (needn't use $\bar{X} \Rightarrow$ by standardisation or 1 st A1) 1 st A1 for correct use of the correct model i.e. awrt 0.035 (allow 0.04 if $P(\bar{X} < 164.6)$ seen) Condone $P(\bar{X} > 164.6) = 0.9652$ or awrt 0.97 <u>only if</u> comparison with 0.95 is made
ALT	Use of z value: Need to see $Z = -1.8(15...)$ and cv of ± 1.6449 (allow 1.64 or better) for 1 st A1
ALT	Use of CR or CV for $\bar{X}$: Need to see " $\bar{X} < 164.7786...$ or CV = ... (awrt 164.8) for 1 st A1 Condone truncation i.e 164.7 or better 2 nd dA1 (dep on M1A1 only) for a correct inference in context. Must mention <u>Mia's belief</u> or <u>mean height of females/women</u> Do NOT award if contradictory statements about hypotheses made e.g. "not sig"
SC	M0 for $\bar{X} \sim N(164.6, ...)$ If they achieve $p =$ awrt 0.035 (o.e. with z -value or CV of 166.3) and a correct conclusion in context is given score M0A0A1 [and SC for awrt 0.97 > 0.95 case]



Qu	Scheme	Marks	AOs
(a)	$\left[P(L < 7.902) = 0.025 \Rightarrow \right] \frac{7.902 - 8}{x} = -1.96$ oe	M1	3.4
	$[x =]0.05^*$	A1cso*	1.1b
	SC B1(mark as M0A1) for $\frac{7.902 - 8}{0.05} = -1.96 \Rightarrow 0.024998$		
		(2)	
(b)	$P(7.94 \leq L \leq 8.09) = 0.8490...$ awrt 0.849	B1	1.1b
		(1)	
(c)	$[P(L < 7.94) =] 0.115069...(awrt 0.115)$ or $[P(L > 8.09) =] 0.03593...(awrt 0.036)$	B1	1.1b
	$[P(L < 7.94) =] 0.115069...(awrt 0.115)$ & $[P(L > 8.09) =] 0.03593...(awrt 0.036)$	B1	1.1b
	Expected income per 500 rods = $\sum (\text{Income} \times \text{probability} \times 500)$ $(500 \times "0.849" \times 0.5) + (500 \times "0.1150..." \times 0.05) + (500 \times "0.03593..." \times 0.4)$ or Expected profit per rod = $\sum (\text{Profit} \times \text{probability})$ $0.30 \times "0.849" + -0.15 \times "0.1150..." + 0.20 \times "0.03593..." [= 0.2446..]$	M1	3.4
	Expected profit per 500 rods $500 \times \sum (\text{Profit} \times \text{probability})$ or $\sum (\text{Income} \times \text{probability} \times 500) - 500 \times 0.2$ $= 500 \times "0.2446..."$ or $= "222.3" - 500 \times 0.2$	M1d	3.1b
	$= [£]122.3...$ awrt [£]122	A1	1.1b
		(5)	
(d)	Let $X \sim B(200, 0.015)$	M1	3.3
	$P(X \leq 5) =$	M1	1.1b
	0.9176...	A1	1.1b
	Manufacturer is unlikely to achieve their aim since $0.9176 < 0.95$	A1ft	2.4
		(4)	

Notes:			(12 marks)
(a)	M1	Using the normal distribution to set up equation. Allow σ for x and awrt ± 1.96	
	A1*	cso For a correct expression for x followed by 0.05 or 0.05000... No incorrect working seen	
(b)	B1	awrt 0.849	
(c)	B1	awrt 0.115 (Implied by awrt 57.5 for number of rods) or awrt 0.036 (Implied by awrt 18 for number of rods)	
	B1	awrt 0.115 (Implied by awrt 57.5 for number of rods) and awrt 0.036 (Implied by awrt 18 for number of rods)	
	M1	Correct method to find the total income of 500 rods. Attempt at all 3 with at least two correct and no extras or Correct method to find sum of all three profits with at least two of 30, -15 or 20 correct. May work in pence but need to be consistent. Allow awrt 24.5 or 0.245	
	M1d	Dep on previous method for finding profit for 500 rods. May work in pence but need to be consistent. Allow $"0.2446..." \times 500$ or "their income" for 500 rods $- 500 \times 0.2$ (accept 499 or 501)	
	A1	All previous marks must be awarded for awrt 122 awrt 12200p NB if uses any integer values for numbers of rods then it is A0 other than for 18 for $L > 8.09$	
(d)	M1	Selecting the appropriate model. May be seen or used. Allow B(200, 0.985) or Po(3) Condone B(0.015, 200) or B(0.985, 200).	
	M1	Writing or using $P(X \leq 5)$ Do not accept $P(X < 6)$ unless found $P(X \leq 5)$	Writing or using $P(X \geq 6)$ Do not accept $P(X > 5)$ unless found $P(X \geq 6)$
	A1	0.92 (Poisson 0.916...)	0.08 or better
	A1ft	Need at least one of the method marks to be awarded. Correct conclusion with the comparison (may be in words). Ft "their $p = 0.9176...$ " as long as $p > 0.9$ If "their $0.9176..." < 0.95$ must ... be unlikely... If "their $0.9176..." > 0.95$ they must say ... be likely... To fit the alternative then $p < 0.1$	